

Assignment:- 4

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Subject: Software Engineering.

⑦ What do you mean by testing tool?

→ A testing tool is a software application that assists in the process of testing by helping testers execute test cases, detect defects, and manage testing activities efficiently. These tools reduce manual effort and improve accuracy.

Key points:

- used to automate or support testing
- Helps in defect tracking and reporting
- Saves time & increases efficiency

Example:

Selenium is used for automation, while JIRA is used for bug tracking.

② Name the tool we used for manual testing.

→ manual testing uses tools mainly for documentation and tracking rather than automation.

Common Tools:

- JIRA - used for defect Tracking
- TestRail - used for managing test cases
- Microsoft Excel / Google Sheets - used for writing test cases.

Example: A tester writes test cases in Excel and logs bugs in JIRA after execution. (1)

③ Explain Software Reliability and Software Quality in detail.

→ ~~Software Reliability~~

- It refers to the ability of a system to perform its intended functions without failure for a specified period under given conditions. It focuses on consistency and stability over time.

Key Points:

- Measures failure-free operation
- Time - dependent concept.
- Indicates system stability

Software quality

- It refers to the degree to which software satisfies user requirements and experience. It is a broader concept that includes various attributes such as functionality, usability, efficiency and security.

Key points:

- Focuses on meeting user needs
- Includes multiple quality attributes
- Not limited to failure-free operation

④ Describe the levels of testing.

Unit testing:

- Unit testing is the process of testing individual components or modules of a software application. It is usually performed by developers during the coding phase.

Key points:

- Tests smallest units of code
- Done by developers
- Helps detect early errors.

Ex: Testing a function that calculates the sum of two numbers.

Integration Testing

- It focuses on testing the interaction between different modules or components. It ensures that combined parts of the system work together correctly.

key points:

- Tests module interaction
- Detects interface errors
- Performed after unit testing

Ex: Testing whether the login module correctly communicates with the database.

System Testing

- It is performed on the complete integrated system to verify that it meets the specified requirements. It checks the system as a whole.

key points:

- Tests entire system
- Validates functional & non-functional requirements
- Performed by testers

Ex: Testing an entire e-commerce application including login, cart and payment.

Acceptance testing:

- It is conducted to determine whether the system satisfies user requirements and is ready for deployment. It is usually performed by clients or end users.

Key Points?

- Final level of testing
- Confirms system readiness
- Includes Alpha and Beta testing.

ex: Client testing the software before approving it for release.

⑤ Explain coding Standards and Guidelines in Software Development.

- They are a set of rules and guidelines followed by developers while writing code to ensure consistency, readability, and maintainability.

Key Points:

- Improve code readability
- Make debugging easier
- Ensure consistency across team
- Help in long-term maintenance

Common Guidelines:

- use meaningful variable names
- maintain proper indentation
- follow naming conventions
- writes comments where necessary
- Avoid unnecessary complexity

ex: using variable name 'total marks' instead of 'tm' improves readability

⑥ Discuss Test case Designer Techniques.

→ Test case design techniques are methods used to create effective test cases that can detect maximum defects with minimum effort.

Important Techniques:-

Equivalence partitioning

- Divides input data into valid and invalid groups
- Reduces number of test cases

Example:

Age input (1-1000): test with 10, 50 (valid)
and -1, 150 (invalid).

Boundary value Analysis

- Focuses on boundary values of input ranges

ex:

For range 1-100 $\rightarrow$ test 0, 1, 100, 101

Decision table testing

- used for complex business logic

ex:

Different combinations of conditions in a login system

State Transition testing

- tests system behaviour for different states

ex:

ATM states: insert card $\rightarrow$ enter PIN $\rightarrow$ withdraw

⑦ Discuss Black Box and white Box testing:

Black Box Testing:

- It is a testing technique where the internal structure of the software is not known. The tester focuses only on inputs and outputs

Key points:

- No knowledge of code required
- Based on requirements
- tests functionality

Unit 22A

Advantages

- Easy to perform
- user-focused testing

Disadvantages

- Limited coverage
- cannot detect internal errors

White Box Testing

- It is a technique where the internal structure and code of the software are known and tested.

Key points:

- Requires programming knowledge
- tests internal logic and code paths
- performed by developers

Advantages

- High coverage
- Detect hidden errors

Disadvantages

- Time-consuming
- Requires skilled testers

Shan
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